

Virtual Dysphoria: A Non-Diagnostic Construct Proposal on Loss of Belonging to the Physical World After Prolonged Immersive Social Virtual Reality Use

André Cicivizzo

Undergraduate Student in Psychology

Universidade Paulista (UNIP)

São Paulo, SP, Brazil

E-mail: andre.cicivizzo@outlook.com

ORCID: <https://orcid.org/0009-0008-8414-8403>

Abstract

Social virtual reality platforms such as VRChat allow users to spend many hours daily embodied as diverse avatars, experiencing gender expression, non-human bodily forms, and community belonging in simulated worlds. Some heavy users report distress when returning to the physical world: the offline environment feels alien, the body mismatched, and the virtual community like home. This article proposes virtual dysphoria as a *non-diagnostic, descriptive construct for this technology-mediated discomfort. Virtual dysphoria is not clinical gender dysphoria and does not pathologize* transgender identity, furry culture, or exploratory avatar use. It names normative subjective distress centered on loss of belonging to the physical world, affective discordance between the self experienced in immersive VR and the offline physical self, and discomfort at the virtual-to-physical transition after prolonged immersive social VR use. Three proposed dimensions are belonging dysphoria (the physical world no longer feels like home), gender-related virtual-physical discordance, and bodily form or species-related virtual-physical discordance. Grounded in research on avatar embodiment, identity disjunction in social VR, depersonalization and derealization after VR exposure, and prolonged immersive use, the proposal distinguishes virtual dysphoria from gender euphoria in affirming contexts, clinical depersonalization disorders, and virtual confabulation. Implications for psychology, platform design, and public health are discussed, along with methodological guidelines for empirical operationalization.

Keywords: virtual dysphoria; virtual reality; VRChat; avatar embodiment; identity; belonging; gender; immersive social VR

1. Introduction

Social virtual reality platforms allow users to meet, interact, and build relationships while embodying customizable avatars in persistent simulated worlds. Among these platforms, VRChat has become particularly visible for the diversity of avatars in use, including feminine, masculine, androgynous, animal, fantasy, and non-human forms. A subset of users engages with such platforms for many hours per day, treating immersive social VR not as occasional entertainment but as a primary site of social life, identity expression, and emotional belonging.

This shift in the scale and function of immersive experience raises questions that remain insufficiently theorized in psychology. When a user spends most of their social and leisure time embodied as an avatar, perceiving, moving, and relating through a simulated body, what happens upon return to the physical world? Anecdotal reports from heavy users describe the offline environment as strange, colorless, or "wrong"; the physical body as mismatched or alien; and the virtual community as the place where they truly belong. These reports suggest a form of distress distinct from memory distortion, although it may co-occur with it.

We propose the term virtual dysphoria to name this phenomenon. The construct is *non-diagnostic

and descriptive: it designates technology-mediated distress arising when prolonged immersive social VR use reorganizes the user's sense of belonging, bodily identity, and existential reference away from the physical world and toward the virtual environment. Virtual dysphoria is not clinical gender dysphoria* and must not be read as a judgment on transgender, furry, or other avatar-based communities. It describes a subjective mismatch experienced during or after the transition from virtual to physical existence, potentially aggravated by intensive immersive use, without equating avatar diversity or identity exploration with pathology.

This article has four objectives: (a) to provide a theoretical foundation for virtual dysphoria; (b) to operationally define the construct and its proposed dimensions; (c) to distinguish it from related clinical and technological concepts; (d) to indicate practical implications and guidelines for empirical investigation.

2. Theoretical Foundation

2.1. Avatar Embodiment and the Proteus Effect

Avatar embodiment refers to the experience of treating a simulated body as one's own during immersive VR. Research on the Proteus effect indicates that avatar appearance can influence self-perception and behavior (Yee & Bailenson, 2007). In fully immersive social VR with six-degrees-of-freedom tracking, embodiment extends beyond visual representation to include proprioceptive and social dimensions: users see their avatar's movements as their own, receive social feedback directed at the avatar, and form relationships while presented as that body.

When embodiment is occasional and brief, avatar effects tend to be transient. When embodiment is prolonged and socially meaningful, the avatar may become a primary reference for self-experience rather than a temporary role.

2.2. Identity Disjunction in Social VR

Kim et al. (2023) studied self-expansion in VRChat across a two-wave panel (N = 486) and found that immersive self-expansion could damage self-esteem and life satisfaction when it produced identity disjunction or self-discrepancy between virtual and offline selves. This work supports the idea that social VR can reorganize identity, with both beneficial and harmful outcomes depending on coherence between virtual and physical self-concepts.

Virtual dysphoria extends this line of reasoning by emphasizing not only self-discrepancy but also *loss of belonging to the physical world and distress upon virtual-to-physical transition*, especially under conditions of heavy daily use.

2.3. Depersonalization, Derealization, and Post-Immersive Distress

VR exposure can induce transient symptoms of depersonalization and derealization (Peckmann et al., 2022; Aardema et al., 2022). Users may report that the physical world feels unreal immediately after sessions. For most users, such effects are short-lived. Virtual dysphoria, as proposed here, is not reducible to transient DPDR symptoms. It involves a sustained reorientation of belonging and bodily reference toward the virtual environment, such that the physical world feels not merely briefly unreal but existentially alien or secondary.

2.4. Gender Expression and Dysphoria in Social VR Contexts

Research on transgender and gender-diverse users in social VR documents both gender euphoria and gender-related distress. Avatar customization and embodiment can affirm internal gender identity and reduce offline gender dysphoria for some users (Reyes & Fisher, 2022; Freeman et al., 2024). Voice-avatar mismatch can also produce gender-related distress in social VR (Li et al., 2024).

Virtual dysphoria incorporates gender-related discordance as one dimension, but defines it specifically

as distress arising from the *contrast between immersive avatar-based gender experience and the offline physical body after prolonged use*, rather than as a replacement for clinical gender dysphoria as defined in diagnostic manuals.

3. Operational Definition of Virtual Dysphoria

We define *virtual dysphoria* as the normative, technology-mediated experience of psychological distress characterized by (a) reduced or lost sense of belonging to the physical world; (b) preference for or emotional dependence on the immersive virtual environment and its community; and (c) affective discordance between characteristics of the embodied avatar (gender, bodily form, species, or persona) and the offline physical body, typically intensified following sessions of prolonged immersive social VR use and upon transition back to physical reality.

Virtual dysphoria is proposed as a *construct describing subjective distress*, not as a diagnostic category. It does not imply that avatar diversity, transgender identity, furry culture, or non-human avatar use is inherently pathological.

3.1. Three Proposed Dimensions

Belonging dysphoria. The user experiences the physical world as alien, insufficient, or no longer "home," while the virtual environment and its community feel like the primary site of belonging. This dimension captures existential displacement rather than mere preference for gaming.

Gender-related virtual-physical discordance. The user experiences distress when transitioning from an avatar whose gender expression aligns more closely with their experienced self to an offline body perceived as mismatched. This may occur in cisgender, transgender, and gender-expansive users alike and is defined by the virtual-physical contrast under immersion, not by pathologizing gender identity itself.

Bodily form and species-related virtual-physical discordance. The user experiences distress when transitioning from a non-human, animal, fantasy, or otherwise non-standard avatar form to the human physical body. This dimension addresses reported discomfort with offline human embodiment after extended habitation of alternative forms in social VR, without reducing such identity practices to delusion or mockery.

3.2. Differential Criteria

We propose the following minimum criteria for identification of virtual dysphoria in research contexts:

1. regular use of immersive social VR (e.g., several hours per day or most days per week);
2. subjective report of distress, alienation, or mismatch upon leaving VR or confronting the physical world;
3. at least one dimension among belonging, gender-related, or form/species-related discordance;
4. temporal association between symptom intensity and immersive use;
5. absence of explanation solely by a primary psychotic or neurological disorder (though comorbidity may exist);
6. user experience framed as distress rather than neutral identity exploration.

4. Mechanisms of Virtual Dysphoria

4.1. Prolonged Embodiment and Habituation

Daily habitation of an avatar increases sensorimotor and social habituation to a simulated body. The physical body becomes comparatively less practiced as a site of self-presentation. Upon headset removal, the user must abruptly reinhabit a body that has become cognitively and affectively secondary.

4.2. Migration of Belonging and Community

Social VR platforms provide friendship, romance, status, ritual, and community belonging. When these bonds are predominantly virtual, offline social space may feel empty or hostile by comparison. Belonging dysphoria arises when the locus of "home" shifts to the simulated world.

4.3. Avatar as Primary Self-Reference

With sufficient immersion, the avatar may function as the user's primary mirror of self: gender, aesthetics, species, voice (including modified voice), and social identity converge on the virtual body. Offline physical traits that contradict this reference become sources of discomfort rather than neutral facts.

4.4. Transition Shock (Virtual-to-Physical)

Virtual dysphoria is often most salient at the **transition point**: removing the headset, seeing one's physical body, hearing one's unmodified voice, and re-entering offline spaces. This transition shock differs from in-session euphoria or in-session safety and helps explain why the same platform can be both affirming and, under heavy use, distress-generating.

4.5. Aggravation Rather Than Pure Causation

Virtual dysphoria does not require that VR create identity or belonging concerns ex nihilo. Pre-existing loneliness, gender questioning, bodily dissatisfaction, or social exclusion may precede heavy VR use. Immersive social VR may **aggravate** displacement by providing a compelling alternative world that intensifies contrast with offline life. This aggravation model preserves individual differences and avoids technological determinism.

5. Distinction From Related Constructs

Virtual confabulation (Cicivizzo, 2026) concerns **memory**: virtual events remembered as physically real. Virtual dysphoria concerns identity and belonging*: the physical world and body feel wrong relative to the virtual self. Both may occur in heavy users of social VR and may be studied as complementary outcomes of prolonged immersion.

6. The VRChat Context

VRChat is frequently cited in community discourse for avatar diversity and long session lengths. Ethnographic work describes users acclimating to extremely long sessions and developing strategies to manage sensory mismatch and phantom sensations (Krell, 2026). Phenomenological accounts from transgender users emphasize VRChat as a site of identity development (Acena, 2023).

Virtual dysphoria is not specific to VRChat alone; it is hypothesized to occur on any immersive social VR platform with comparable embodiment and community persistence. VRChat serves as a paradigmatic case because of avatar diversity (including non-human and cross-gender forms), social density, and documented heavy-use subcultures.

7. Implications

7.1. Clinical and Counseling Psychology

Clinicians may encounter clients who report that offline life feels unreal or unbearable while virtual life feels authentic. Virtual dysphoria offers a non-moralizing framework for discussing such experiences without dismissing transgender or avatar-based identity exploration. Assessment should distinguish affirming identity development from distress associated with disproportionate immersion and loss of offline belonging.

7.2. Public Health and Prevention

If empirical research confirms dose-response relationships between immersive hours and virtual

dysphoria, public health messaging might address balanced use, offline anchoring practices, and transition rituals after sessions. This is not an argument against VR, but for informed use.

7.3. Platform Design and Ethics

Platforms benefiting from engagement may have structural incentives toward prolonged immersion. Ethical design could include session awareness tools, offline reintegration prompts, and community norms supporting healthy transitions. Research on virtual dysphoria may inform such interventions.

7.4. Relationship to Virtual Confabulation

Users experiencing belonging displacement may also show memory source confusion between virtual and physical events. Integrated models of *immersive self disruption* could study virtual dysphoria and virtual confabulation as joint outcomes of the same exposure profile, sharing immersion intensity as a predictor while remaining conceptually distinct.

8. Methodological Proposal for Empirical Investigation

Operationalization of virtual dysphoria requires self-report instruments and, ideally, longitudinal designs tracking immersive hours and transition distress.

8.1. Suggested Design

A correlational study comparing *heavy users (e.g., 2 or more hours/day, 4 or more days/week on social VR) with light users* (under 5 hours/week). Measures administered online via structured questionnaire after CEP approval.

8.2. Proposed Measures

Dependent variables: Virtual Dysphoria Scale (to be developed); subscales for belonging, gender-related, and form/species-related discordance.

Independent variables: weekly immersive hours; session length; months of regular use; avatar type (human vs non-human; same vs different gender presentation offline).

Covariates: offline social support; clinical gender dysphoria screens (to separate constructs); depersonalization scales; problematic social VR use.

8.3. Hypotheses

H1: Heavy immersive social VR use will predict higher virtual dysphoria scores than light use.

H2: Belonging dysphoria will correlate positively with daily immersive hours.

H3: Users of avatars with marked gender or form discrepancy from offline presentation will report higher virtual-physical discordance, moderated by immersion intensity.

H4: Virtual dysphoria will correlate positively with, but remain distinct from, virtual confabulation scores.

9. Limitations and Research Agenda

The construct proposed here is predominantly theoretical. Existing literature documents related phenomena but not virtual dysphoria as an integrative construct. Key open questions include: Does virtual dysphoria persist between sessions or peak only at transition? Do adolescents show greater susceptibility during identity formation? Can offline anchoring interventions reduce belonging dysphoria without harming affirming identity exploration? How do species- or form-related avatar users conceptualize distress in their own terms?

Future work should prioritize scale development, cross-cultural validation, and longitudinal designs.

Qualitative interviews with heavy VRChat users should inform item wording and avoid pathologizing community identity practices.

10. Final Considerations

Prolonged immersive social VR can reorganize where users feel they belong, which body they experience as "theirs," and which world feels real. Virtual dysphoria, as proposed here, names the distress that may arise when the physical world and body cease to feel like home after extended life in simulated embodiment. The construct is offered as a tool for research and clinical conversation, not as stigma toward avatar diversity or identity exploration.

Understood alongside virtual confabulation, it suggests that immersive technologies may affect not only what users remember, but where and as whom they feel they exist. As social VR continues to expand, psychology requires vocabulary adequate to these experiences. Virtual dysphoria is proposed as a contribution to that vocabulary.

Author Contributions

André Cicivizzo: conceptualization, literature review, writing (original draft), writing (review and editing).

Conflict of Interest

The author declares no conflict of interest.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Ethics Statement

This is a theoretical article. No human or animal participants were involved. Therefore, approval from a research ethics committee was not required.

Data Availability Statement

Not applicable. No datasets were generated or analyzed in this theoretical article.

References

Aardema, F., et al. (2022). Experiences of depersonalization/derealization among users of virtual reality applications: A cross-sectional survey. *Cyberpsychology, Behavior, and Social Networking*, 25(8), 495-502.

Acena, D. (2023). Gender identity development in social virtual reality: A phenomenology [Doctoral dissertation, University of Northern Colorado].

Cicivizzo, A. (2026). Virtual confabulation: A construct proposal on the integration of immersive episodic memories into the autobiographical record. PsyArXiv. https://osf.io/preprints/psyarxiv/hrnxe_v1

Freeman, G., et al. (2024). Gender expression and gender identity in virtual reality: Avatars, role-adoption, and social interaction in VRChat. *Frontiers in Virtual Reality*, 5, 1305758.

Kim, S., et al. (2023). The double-edged influence of self-expansion in the metaverse: A two-wave

panel assessment of identity perception, self-esteem, and life satisfaction. *Computers in Human Behavior*, 149, 107984.

Krell, F. (2026). Becoming a social VR user: Acculturated perception and social behavior in long-term VRChat communities. *Presence: Virtual and Augmented Reality*.

Li, Y., et al. (2024). Springboard, roadblock or "crutch"? How transgender users leverage voice changers for gender presentation in social virtual reality. *arXiv*. <https://doi.org/10.48550/arxiv.2402.08217>

Peckmann, C., et al. (2022). Virtual reality induces symptoms of depersonalization and derealization: A longitudinal randomised control trial. *Computers in Human Behavior*, 131, 107233.

Reyes, A., & Fisher, K. (2022). Gender identity development in social virtual reality. In *Proceedings of CHI Conference on Human Factors in Computing Systems*.

Yee, N., & Bailenson, J. (2007). The Proteus effect: The effect of transformed self-representation on behavior. *Human Communication Research*, 33(3), 271-290.